

Assignment 1

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(1)

Since $X = [X_1|X_2]$,

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = (X'X)^{-1}X'y = \begin{pmatrix} X_1'X_1 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{pmatrix}^{-1} \begin{pmatrix} X_1'y \\ X_2'y \end{pmatrix}.$$

Let $M_i = I - X_i(X_i'X_i)^{-1}X_i'$. Then $X_1'X_1 - X_1'X_2(X_2'X_2)^{-1}X_2'X_1 = X_1'M_2X_1$ and likewise for $X_2'M_1X_2$, so by the partitioned-inverse formula (3.71)

$$(X'X)^{-1} = \begin{pmatrix} (X_1'M_2X_1)^{-1} & -(X_1'M_2X_1)^{-1}X_1'X_2(X_2'X_2)^{-1} \\ -(X_2'M_1X_2)^{-1}X_2'X_1(X_1'X_1)^{-1} & (X_2'M_1X_2)^{-1} \end{pmatrix}.$$

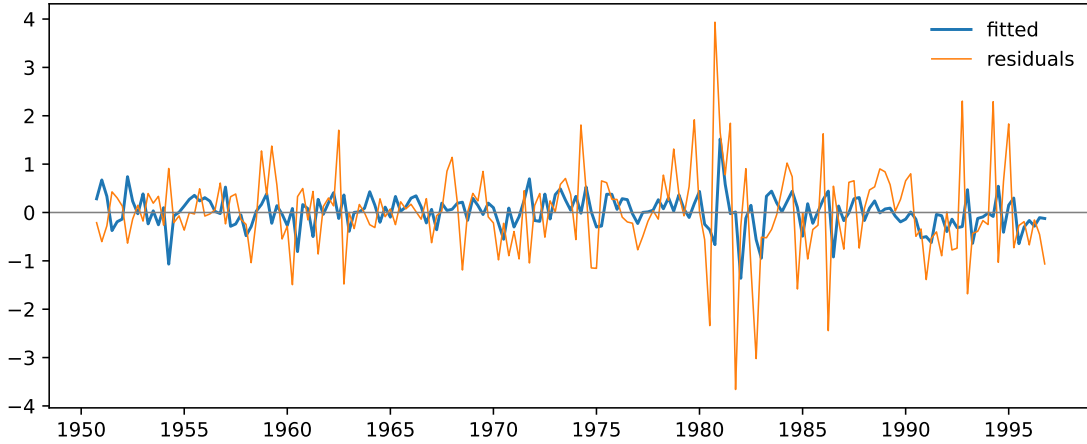
Multiplying out each row:

$$\begin{aligned} b_1 &= (X_1'M_2X_1)^{-1}X_1'(I - X_2(X_2'X_2)^{-1}X_2')y = (X_1'M_2X_1)^{-1}X_1'M_2y, \\ b_2 &= (X_2'M_1X_2)^{-1}X_2'(I - X_1(X_1'X_1)^{-1}X_1')y = (X_2'M_1X_2)^{-1}X_2'M_1y. \end{aligned}$$

(2)

Sample 1950:4–1996:4, $n = 185$. OLS gives (standard errors in brackets)

$$\widehat{\Delta r}_t = \underset{(0.1256)}{-0.2319} + \underset{(0.0200)}{0.0161} \pi_{t-1} + \underset{(5.7589)}{18.3807} \Delta y_{t-1} + \underset{(0.0741)}{0.2375} \Delta r_{t-1} - \underset{(0.0725)}{0.1540} \Delta r_{t-2}, \quad R^2 = 0.134.$$



Fitted values and residuals, 1950:4–1996:4.

Residuals on a constant and $\hat{y}$: both coefficients are zero (up to 10^{-16} rounding) and $R^2 = 0$.

Fitted values on a constant and e : the slope is zero and $R^2 = 0$; the intercept is 0.0134, the mean of Δr_t .

(3)

Let $X_1 = [\iota, \Delta y_{t-1}, \Delta r_{t-1}, \Delta r_{t-2}]$ and $x_2 = \pi_{t-1}$. Then $e_t = M_1 \Delta r$, $e_t^\pi = M_1 x_2$, and regressing e_t on e_t^π gives

$$\hat{\beta} = (x_2' M_1 x_2)^{-1} x_2' M_1 \Delta r = 0.0161,$$

the same as the coefficient on π_{t-1} in (1). The residuals are identical to the residuals from (1) (largest difference 9×10^{-15}).

(4)

Sample 1953:1–1996:4, $n = 176$, with $c_t = \log CE_t$ and $y_t = \log YD_t$. OLS gives

$$\widehat{\Delta c}_t = \underset{(0.0012)}{0.0051} + \underset{(0.0545)}{0.3058} \Delta y_t + \underset{(0.0544)}{0.1242} \Delta y_{t-1} + \underset{(0.0551)}{0.0425} \Delta y_{t-2} + \underset{(0.0545)}{0.1239} \Delta y_{t-3} - \underset{(0.0539)}{0.1233} \Delta y_{t-4}.$$

Method 1. Let $w = (0, 1, 1, 1, 1, 1)'$. Then

$$\hat{\gamma} = w' \hat{\beta} = 0.4732, \quad \text{se}(\hat{\gamma}) = \sqrt{w' \widehat{\text{Var}}(\hat{\beta}) w} = 0.1000.$$

Method 2. Substitute $\beta_2 = \gamma - \beta_3 - \beta_4 - \beta_5 - \beta_6$ into (2):

$$\Delta c_t = \beta_1 + \gamma \Delta y_t + \beta_3(\Delta y_{t-1} - \Delta y_t) + \beta_4(\Delta y_{t-2} - \Delta y_t) + \beta_5(\Delta y_{t-3} - \Delta y_t) + \beta_6(\Delta y_{t-4} - \Delta y_t) + \text{errors}.$$

The coefficient on Δy_t is now γ . OLS gives $\hat{\gamma} = 0.4732$ with standard error 0.1000.

(5)

Original regression: $\hat{\beta}_2 = (X_2' M_1 X_2)^{-1} X_2' M_1 y$, residuals $\hat{u} = M_X y$, and $y = X_1 \hat{\beta}_1 + X_2 \hat{\beta}_2 + M_X y$. Facts used: $P_1 X_1 = X_1$, $M_1 X_1 = 0$, $P_X X_i = X_i$, $M_X X_i = 0$, $M_1 M_X = M_X$, $P_1 P_X = P_1$; all projections are symmetric and idempotent.

(a) $y = X_2 \beta_2 + u$.

$$\begin{aligned} \tilde{\beta}_2 &= (X_2' X_2)^{-1} X_2' y \neq \hat{\beta}_2, \\ \tilde{u} &= y - X_2 \tilde{\beta}_2 = M_2 y \neq M_X y. \end{aligned}$$

β_2 : not the same. Residuals: not the same.

(b) $P_1 y = X_2 \beta_2 + u$.

$$\begin{aligned} \tilde{\beta}_2 &= (X_2' X_2)^{-1} X_2' P_1 y \neq \hat{\beta}_2, \\ \tilde{u} &= P_1 y - X_2 \tilde{\beta}_2 = M_2 P_1 y \neq M_X y. \end{aligned}$$

β_2 : not the same. Residuals: not the same.

(c) $P_1 y = P_1 X_2 \beta_2 + u$.

$$\begin{aligned} \tilde{\beta}_2 &= (X_2' P_1 P_1 X_2)^{-1} X_2' P_1 P_1 y = (X_2' P_1 X_2)^{-1} X_2' P_1 y \neq \hat{\beta}_2, \\ \tilde{u} &= P_1 y - P_1 X_2 \tilde{\beta}_2 = P_1 (y - X_2 \tilde{\beta}_2) \neq M_X y. \end{aligned}$$

β_2 : not the same. Residuals: not the same.

(d) $P_X y = X_1 \beta_1 + X_2 \beta_2 + u.$

$$\begin{aligned}\tilde{\beta} &= (X'X)^{-1} X' P_X y = (X'X)^{-1} (P_X X)' y = (X'X)^{-1} X' y = \hat{\beta} \Rightarrow \tilde{\beta}_2 = \hat{\beta}_2, \\ \tilde{u} &= P_X y - X \hat{\beta} = P_X y - P_X y = 0 \neq M_X y.\end{aligned}$$

β_2 : the same. Residuals: not the same.

(e) $P_X y = X_2 \beta_2 + u.$

$$\begin{aligned}\tilde{\beta}_2 &= (X_2' X_2)^{-1} X_2' P_X y = (X_2' X_2)^{-1} (P_X X_2)' y = (X_2' X_2)^{-1} X_2' y \neq \hat{\beta}_2, \\ \tilde{u} &= P_X y - X_2 \tilde{\beta}_2 = M_2 P_X y \neq M_X y.\end{aligned}$$

β_2 : not the same (same as (a)). Residuals: not the same.

(f) $M_1 y = X_2 \beta_2 + u.$

$$\begin{aligned}\tilde{\beta}_2 &= (X_2' X_2)^{-1} X_2' M_1 y \neq (X_2' M_1 X_2)^{-1} X_2' M_1 y = \hat{\beta}_2, \\ \tilde{u} &= M_1 y - X_2 \tilde{\beta}_2 = M_2 M_1 y \neq M_X y.\end{aligned}$$

β_2 : not the same. Residuals: not the same.

(g) $M_1 y = M_1 X_2 \beta_2 + u.$

$$\begin{aligned}\tilde{\beta}_2 &= (X_2' M_1 M_1 X_2)^{-1} X_2' M_1 M_1 y = (X_2' M_1 X_2)^{-1} X_2' M_1 y = \hat{\beta}_2, \\ \tilde{u} &= M_1 y - M_1 X_2 \hat{\beta}_2 = M_1 (X_1 \hat{\beta}_1 + X_2 \hat{\beta}_2 + M_X y) - M_1 X_2 \hat{\beta}_2 = M_1 M_X y = M_X y.\end{aligned}$$

β_2 : the same. Residuals: the same.

(h) $M_1 y = X_1 \beta_1 + M_1 X_2 \beta_2 + u.$ Since $X_1' M_1 X_2 = (M_1 X_1)' X_2 = 0$, the cross-product matrix is block diagonal, so

$$\begin{aligned}\tilde{\beta}_1 &= (X_1' X_1)^{-1} X_1' M_1 y = 0, \\ \tilde{\beta}_2 &= (X_2' M_1 X_2)^{-1} X_2' M_1 y = \hat{\beta}_2, \\ \tilde{u} &= M_1 y - X_1 \cdot 0 - M_1 X_2 \hat{\beta}_2 = M_X y \quad (\text{as in (g)}).\end{aligned}$$

β_2 : the same. Residuals: the same.

(i) $M_1 y = M_1 X_1 \beta_1 + M_1 X_2 \beta_2 + u.$ Since $M_1 X_1 = 0$, the X_1 block is all zeros and β_1 is not identified; the regression reduces to (g):

$$\begin{aligned}\tilde{\beta}_2 &= (X_2' M_1 X_2)^{-1} X_2' M_1 y = \hat{\beta}_2, \\ \tilde{u} &= M_1 y - M_1 X_2 \hat{\beta}_2 = M_X y.\end{aligned}$$

β_2 : the same. Residuals: the same.

(j) $P_X y = M_1 X_2 \beta_2 + u$. Here $M_1 P_X y = M_1 (X_1 \hat{\beta}_1 + X_2 \hat{\beta}_2) = M_1 X_2 \hat{\beta}_2$, so

$$\tilde{\beta}_2 = (X_2' M_1 X_2)^{-1} X_2' M_1 P_X y = (X_2' M_1 X_2)^{-1} X_2' M_1 X_2 \hat{\beta}_2 = \hat{\beta}_2,$$

$$\tilde{u} = P_X y - M_1 X_2 \hat{\beta}_2 = X_1 \hat{\beta}_1 + P_1 X_2 \hat{\beta}_2 = P_1 (X_1 \hat{\beta}_1 + X_2 \hat{\beta}_2) = P_1 P_X y = P_1 y \neq M_X y.$$

β_2 : the same. Residuals: not the same.

β_2 is the same in (d), (g), (h), (i), (j). The residuals are the same only in (g), (h), (i).